AQ2.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

1 point
Match the matrices in Column A with their inverses in Column B.

- [ ] a → → iii
- [ ] a → → ii
- [ ] b → → iii
- [ ] b → → i
- [ ] c → → i
- [ ] c → → ii

No, the answer is incorrect.
Score: 0
Accepted Answers:
a → → ii
b → → iii
c → → i

(Use the below information to answer questions 2 and 3).
Consider a system of linear equations

$$\begin{aligned} 2x_1 - x_2 &= 3 \\ x_1 - x_3 &= 3 \\ x_2 - x_3 &= 2 \end{aligned}$$ 2x1 − x2 = 3x1 − x3 = 3x2 − x3 = 2

Let the matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$A = [2−1 0 1 0 −1 0 1 −1], $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], and $b = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$b = [332].

1 point
Choose the set of correct options.

- [ ] $A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$A−1 = [2−1 0 −1 1 0 0 −1]
- [ ]

A^{-1} = $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$A−1 = $\begin{bmatrix} & & \\ & & \\ 1 & 1 & 1 \\ -1 & -2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

☐

Adjoint of the matrix AA is $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$ $\begin{bmatrix} & & \\ & & \\ 1 & 1 & 1 \\ -1 & -2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

☐

$det(A)$ = 1det(A) = 1.

No, the answer is incorrect.
Score: 0
Accepted Answers:

A^{-1} = $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$A−1 = $\begin{bmatrix} & & \\ & & \\ 1 & 1 & 1 \\ -1 & -2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

Adjoint of the matrix AA is $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & 2 \\ 1 & -2 & 1 \end{bmatrix}$ $\begin{bmatrix} & & \\ & & \\ 1 & 1 & 1 \\ -1 & -2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$

$det(A)$ = 1det(A) = 1.

1 point

Choose the set of correct options.

☐

x_1 = − 2x1 = −2.

☐

x_2 = 1x2 = 1.

☐

x_3 = − 1x3 = −1.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:

x_2 = 1x2 = 1.

x_3 = − 1x3 = −1.

1 point

Choose the set of correct options.

☐

A system of linear equations $Ax = b$Ax = b is called a homogeneous system of linear equations if $b \neq 0$b = 0.

☐

A system of linear equations $Ax = b$Ax = b is called a non-homogeneous system of linear equations if $b \neq 0$b = 0

☐

If vv is a solution of the system of linear equations $Ax = b$Ax = b, then $\frac{1}{c}v$c1v is a solution of system of linear equations $cAx = b$cAx = b, where $c \neq 0$c = 0.

☐

Let $Ax = b$Ax = b be a system of linear equations. If AA is invertible, then $adj(A)x = b$adj(A)x = b also has a solution.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A system of linear equations $Ax = b$Ax = b is called a non-homogeneous system of linear equations if $b \neq 0$b = 0
If vv is a solution of the system of linear equations $Ax = b$Ax = b, then $\frac{1}{c}v$c1v is a solution of system of linear equations $cAx = b$cAx = b, where $c \neq 0$c = 0.
Let $Ax = b$Ax = b be a system of linear equations. If AA is invertible, then $adj(A)x = b$adj(A)x = b also has a solution.

Suppose CC is the cofactor matrix of $\begin{bmatrix} 3 & 2 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Let C_{ij} Cij be the ijij-th element of the cofactor matrix CC. Answer the questions 5,6 and 7.

Find the value of C_{12} C12.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
Find the value of C_{22} C22.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -3
1 point
Find the sum of the elements in the third row of CC.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Level 2:

1 point
Which of the following square matrices of order 3 are the same as their adjoint matrices?
- ☐ Identity matrix.
- ☐ Zero matrix.
- ☐ Any scalar matrix.

☐ Any diagonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Identity matrix.

Zero matrix.

1 point

Choose the set of correct options.

[Hint: For the first option, find out the adjoint matrix of an arbitrary square upper triangular matrix of order 22, i.e., take the matrix $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$[a0bd] and find out its adjoint matrix.]

☐ The adjoint of a $2 \times 2$2 × 2 real upper triangular matrix is an upper triangular matrix.

☐ There exists a square matrix of order 33, such that $A = A^{-1} = adj(A)$A = A−1 = adj(A).

☐ If $A = A^{-1}$A = A−1, then $det(A)$det(A) must be 11.

☐ If $A = A^{-1}$A = A−1, then AA must be an identity matrix.

☐ If $A = A^{-1}$A = A−1, then $A^2 = I$A2 = I.

☐ If $A^{-1} = adj(A)$A−1 = adj(A), then $det(A)$det(A) must be 11.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The adjoint of a $2 \times 2$2 × 2 real upper triangular matrix is an upper triangular matrix.

There exists a square matrix of order 33, such that $A = A^{-1} = adj(A)$A = A−1 = adj(A).

If $A = A^{-1}$A = A−1, then $A^2 = I$A2 = I.

If $A^{-1} = adj(A)$A−1 = adj(A), then $det(A)$det(A) must be 11.

1 point

If A and BA and B are squares matrices of order 22, then choose the set of correct options.

[Hint: Find out the adjoint matrix of an arbitrary square matrix of order 22, i.e., take the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$[ac bd] and find out its adjoint matrix.]

☐ $adj(AB) = adj(A)adj(B)$adj(AB) = adj(A)adj(B).

☐ $adj(AB) = adj(B)adj(A)$adj(AB) = adj(B)adj(A).

☐ $adj(A + B) = adj(A) + adj(B)$adj(A + B) = adj(A) + adj(B).

☐ $adj(A^T) = adj(A)^T$ adj(AT) = adj(A)T .

☐ $adj(A^{-1}) = adj(A)^{-1}$ adj(A−1) = adj(A)−1.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$adj(AB) = adj(B)adj(A)$adj(AB) = adj(B)adj(A).

$adj(A + B) = adj(A) + adj(B)$adj(A + B) = adj(A) + adj(B).

$adj(A^T) = adj(A)^T$ adj(AT) = adj(A)T .
$adj(A^{-1}) = adj(A)^{-1}$ adj(A−1) = adj(A)−1.

Check Answers
Your score is: 0/10